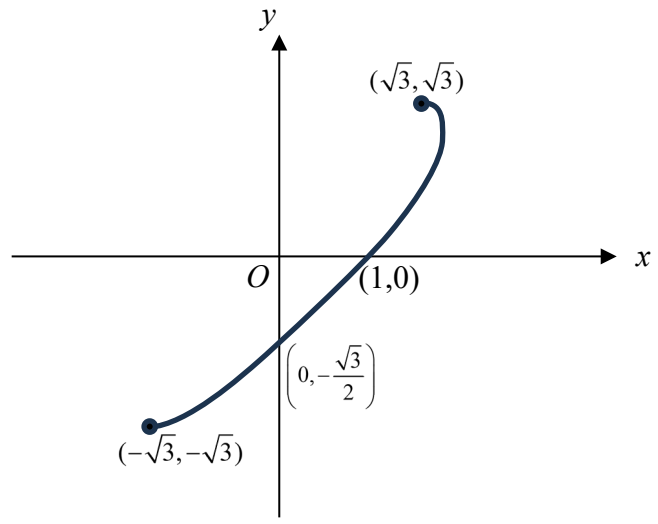
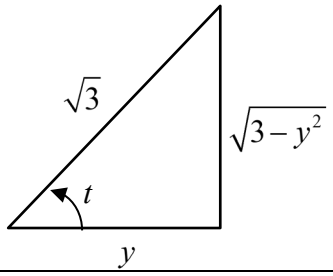
DHS 2026 Prelim H2 Math Paper 2 Suggested Solutions

Qn	Suggested Solution
1(a)	$\dfrac{2x^2-3x+5}{x^2+3x-4}>1$ $\dfrac{2x^2-3x+5-(x^2+3x-4)}{x^2+3x-4}>0$ $\dfrac{x^2-6x+9}{x^2+3x-4}>0$ $\dfrac{(x-3)^2}{(x-1)(x+4)}>0, \quad x\neq -4 \text{ or } 1$ Since $(x-3)^2>0$ for all $x\in\mathbb{R}\setminus\{3\}$, $(x-1)(x+4)>0, \ x\neq 3$ $\therefore x<-4$ or $x>1, x\neq 3$ [Alt : $1<x<3$ or $x>3$]
1(b)	For $x\neq 0$, Replace x by $\dfrac{1}{x}$ in $\dfrac{2x^2-3x+5}{x^2+3x-4}>1$ $\dfrac{2\left(\frac{1}{x}\right)^2-3\left(\frac{1}{x}\right)+5}{\left(\frac{1}{x}\right)^2+3\left(\frac{1}{x}\right)-4}>1$ $\dfrac{\frac{2}{x^2}-\frac{3}{x}+5}{\frac{1}{x^2}+\frac{3}{x}-4}>1$ $\dfrac{2-3x+5x^2}{1+3x-4x^2}>1$ Replace x by $\dfrac{1}{x}$ in the previous solution: $\dfrac{1}{x}<-4$ or $\dfrac{1}{x}>1 \ (\dfrac{1}{x}\neq 3)$ $-\dfrac{1}{4}<x<1, \ x\neq\dfrac{1}{3}$ $\left[\text{Alt}: -\dfrac{1}{4}<x<\dfrac{1}{3} \text{ or } \dfrac{1}{3}<x<1\right]$ Note that $x=0$ is a solution as $\dfrac{2-3(0)+5(0)^2}{1+3(0)-4(0)^2}=2>1.$

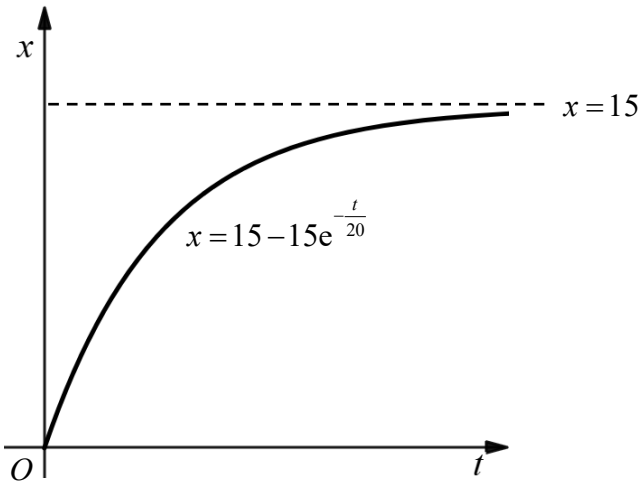
Qn	Suggested Solution
2(a)	Area of a parallelogram formed by sides OA and OB or twice the area of the triangle OAB.
2(b)	$3\mathbf{r}\times(\mathbf{b}-\mathbf{a})=\mathbf{0}$ $3\mathbf{r}\,//\,\mathbf{b}-\mathbf{a}\Rightarrow 3\mathbf{r}=\lambda(\mathbf{b}-\mathbf{a}),\ \lambda\in\mathbb{R}\Rightarrow \mathbf{r}=\frac{\lambda}{3}(\mathbf{b}-\mathbf{a})$ Point R lies on the **line** passing through origin O and parallel to $\overrightarrow{AB}$.
2(c)	$\mathbf{r}\cdot\mathbf{b}=\mathbf{r}\cdot\mathbf{a}$ $\mathbf{r}\cdot(\mathbf{b}-\mathbf{a})=0$ Point R lies on the **plane** containing the origin O with normal $\overrightarrow{AB}$.
2(d)	Let θ be the required acute angle. $\cos\theta = \frac{\left\lvert\begin{pmatrix}-3\\-6\\2\end{pmatrix}\bullet\begin{pmatrix}1\\0\\0\end{pmatrix}\right\rvert}{\left\lvert\sqrt{49}\right\rvert\left\lvert\sqrt{1}\right\rvert}=\frac{\lvert-3\rvert}{7}=\frac{3}{7}$ Using the trigonometric identity, $\sin^2\theta+\cos^2\theta=1$ $\sin\theta=\sqrt{1-\cos^2\theta}$ $=\sqrt{1-\left(\frac{3}{7}\right)^2}$ $=\frac{2\sqrt{10}}{7}$ <u>Alternatively</u> Let θ be the required acute angle. $\sin\theta = \frac{\left\lvert\begin{pmatrix}-3\\-6\\2\end{pmatrix}\times\begin{pmatrix}1\\0\\0\end{pmatrix}\right\rvert}{\left\lvert\sqrt{49}\right\rvert\left\lvert\sqrt{1}\right\rvert}=\frac{\left\lvert\begin{pmatrix}0\\2\\6\end{pmatrix}\right\rvert}{7}$ $=\frac{2\sqrt{10}}{7}$

Qn	Suggested Solution
3(a)	Substitute $y = -x + k$ into $(x + y)^2 + axy = 6$, $(x - x + k)^2 + ax(-x + k) = 6$ $k^2 - ax^2 + akx - 6 = 0$ $ax^2 - akx + (6 - k^2) = 0$ (shown) ....(1)
3(b)	For (1) to have real roots, Discriminant ≥ 0 $a^2k^2 - 4a(6 - k^2) \geq 0$ $(a + 4)k^2 - 24 \geq 0$ $k^2 - \frac{24}{a+4} \geq 0$ ...(2) $\lvert k \rvert \geq 2$ $k \leq -2$ or $k \geq 2$ $(k - 2)(k + 2) \geq 0$ $k^2 - 4 \geq 0$ ...(3) Compare (2) & (3) : $\frac{24}{a+4} = 4$ $24 = 4a + 16$ $\therefore a = 2$

Qn	Suggested Solution
4(a)	$\sin t + \sqrt{3}\cos t = R\sin(t+\alpha)$ $= R\left[\sin t\cos\alpha + \cos t\sin\alpha\right]$ Comparing, $R\cos\alpha = 1$ --- (1) $R\sin\alpha = \sqrt{3}$ --- (2) $\therefore R = \sqrt{\left(\sqrt{3}\right)^2 + 1^2} = 2$ and $\tan\alpha = \sqrt{3} \Rightarrow \alpha = \frac{\pi}{3}$ $\therefore x = 2\sin\left(t + \frac{\pi}{3}\right)$
4(b)	By observation from **(a)**, maximum $x = 2$ [This occurs when $\sin\left(t+\frac{\pi}{3}\right) = 1$ i.e. $t + \frac{\pi}{3} = \frac{\pi}{2} \Rightarrow t = \frac{\pi}{6}$ which satisfy $0 \le t \le \pi$.] If minimum $x = -R = -2$, $\sin\left(t+\frac{\pi}{3}\right) = -1$. Considering the nearest possible solutions to $0 \le t \le \pi$, we have $t + \frac{\pi}{3} = -\frac{\pi}{2}$ or $\frac{3\pi}{2} \Rightarrow t = -\frac{5\pi}{6}$ or $\frac{7\pi}{6}$, both of which are not within $0 \le t \le \pi$. Thus, the minimum value of x is not $-R$. **<u>Alternative</u>** $0 \le t \le \pi \Rightarrow \frac{\pi}{3} \le t + \frac{\pi}{3} \le \frac{4\pi}{3} \Rightarrow -\frac{\sqrt{3}}{2} \le \sin\left(t+\frac{\pi}{3}\right) \le 1$ $\Rightarrow -\sqrt{3} \le 2\sin\left(t+\frac{\pi}{3}\right) \le 2$ Hence, min value of x is $-\sqrt{3} \ne -2 = R$.
4(c)	 When $y = 0$, $\sqrt{3}\cos t = 0 \Rightarrow t = \frac{1}{2}\pi$ $\Rightarrow x = \sin\left(\frac{1}{2}\pi\right) + \sqrt{3}\cos\left(\frac{1}{2}\pi\right) = 1$

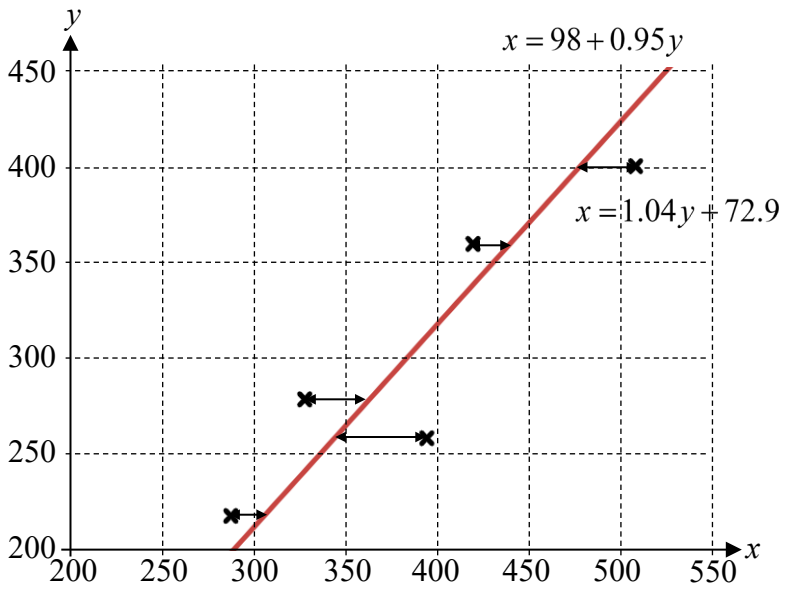
	When $x = 0, \sin t + \sqrt{3}\cos t = 0$ $\sin t = -\sqrt{3}\cos t \Rightarrow \tan t = -\sqrt{3} \Rightarrow t = \frac{2\pi}{3}$ $\Rightarrow y = \sqrt{3}\left(-\frac{1}{2}\right) = -\frac{\sqrt{3}}{2}$
4(d)	$y = \sqrt{3}\cos t \Rightarrow \cos t = \frac{y}{\sqrt{3}}$ $\sin t = \frac{\sqrt{3-y^2}}{\sqrt{3}}$ $x = \frac{\sqrt{3-y^2}}{\sqrt{3}} + y, -\sqrt{3} \le y \le \sqrt{3}.$ 

Qn	Suggested Solution
5(a)	Rate in $= 0.075 \times 10$ $= \frac{3}{4}$ g/min Rate out $= \frac{x}{200} \times 10$ $= \frac{1}{20}x$ g/min $\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{3}{4} - \frac{1}{20}x.$
5(b)	$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{3}{4} - \frac{1}{20}x$ $\frac{20}{15-x}\frac{\mathrm{d}x}{\mathrm{d}t} = 1$ Integrate w.r.t t, $\int \frac{20}{15-x}\mathrm{d}x = \int 1\,\mathrm{d}t$ $-20\ln\lvert 15-x\rvert = t + c$ $\lvert 15-x\rvert = \mathrm{e}^{-\frac{t}{20}-\frac{c}{20}}$ $15-x = A\mathrm{e}^{-\frac{t}{20}}, A = \pm\mathrm{e}^{-\frac{c}{20}}$ $x = 15 - A\mathrm{e}^{-\frac{t}{20}}$ *Remark: If student attempts to remove modulus by reason, they should explain that the concentration of nitrogen in the tank at any one point cannot be greater than the concentration of the nitrogen from the nutrient solution pumped in.* $\frac{x}{200} < 0.075$ $x < 15$ $15 - x > 0$ *or* *Remark: Since the tank initially contains pure water and the more concentrated nitrogen from the nutrient solution is being pumped into the tank, the amount of nitrogen in the tank should always be increasing.* $\frac{3}{4} - \frac{x}{20} > 0$ $\frac{3}{4} > \frac{x}{20}$ $15 > x$ $15 - x > 0$

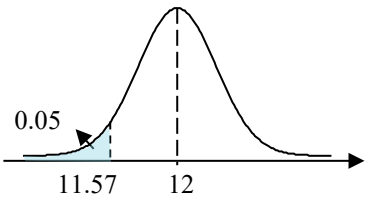
	Sub $t = 0,\ x = 0,$ $0 = 15 - A$ $A = 15$ $x = 15 - 15\mathrm{e}^{-\frac{t}{20}}$
5(c)	 The nitrogen mass will stabilize at 15 grams in the long run.

Qn	Suggested Solution
6(a)	Total number of ornaments that are either white or with a moon pictorial design (or both) but not green $=12+2+4+5+11=34$ Total number of ornaments that are not green = 90 – (13+10+8) = 59 **Method 1: Using reduced sample space** Required Probability $=\frac{34}{59}\times\frac{33}{58}$ $\left(\text{or}\frac{{}^{34}C_2}{{}^{59}C_2}\right)=\frac{561}{1711}$ (or 0.328 (to 3 s.f.)) **Method 2: Using conditional probability formula** Required Probability $=\frac{\frac{34}{90}\times\frac{33}{89}}{\frac{59}{90}\times\frac{58}{89}}$ $\left(\text{or}\frac{\frac{{}^{34}C_2}{{}^{90}C_2}}{\frac{{}^{59}C_2}{{}^{90}C_2}}\right)=\frac{561}{1711}$ (or 0.328 (to 3 s.f.))
6(b)	Number of blue stars = 6 Total number of blue ornaments that are not star = 3 + 5 =8 Total number of star ornaments that are not blue = 12 + 9 + 13 =34 Total number of ornaments that are not star and not blue = 90 – 6 – 8 – 34 = 42 Required Probability = P(one blue ornament that is not a star and one star ornament that is not blue) + P(one blue star and one ornament that is not a star and not blue) $=\frac{8}{90}\times\frac{34}{89}\times 2$ $\left(\text{or}\frac{{}^{8}C_1\,{}^{34}C_1}{{}^{90}C_2}\right)$ + $\frac{6}{90}\times\frac{42}{89}\times 2$ $\left(\text{or}\frac{{}^{6}C_1\,{}^{42}C_1}{{}^{90}C_2}\right)$ $=\frac{524}{4005}$ (or 0.131 (to 3sf))
6(c)	Let x be the number of ornaments of his favourite type. $\frac{x}{90}\times\frac{(x-1)}{89}$ $\left(\text{or}\frac{{}^{x}C_2}{{}^{90}C_2}\right)=\frac{28}{4005}$ Solving (algebraically or using GC), $x=8\Rightarrow$ favourite ornament : Green Moon

Qn	Suggested Solution
7(a)	Case 1 - 4 "S": 1 way Case 2 - 3 "S": ${}^5C_1$ ways Case 3 – 2 "S" or 2 "E" or both: ${}^2C_1 \times {}^5C_2 + 1 = 21$ ways Case 4 – distinct letters: ${}^6C_4$ ways Number of selections without restrictions $= 1 + 5 + 21 + 15 = 42$
7(b)(i)	Number of ways to arrange "AMNT" as one unit and 6 other letters $= \frac{7!}{4!2!}$ Number of ways to arrange "AMNT" $= 4!$ Number of arrangements required $= \frac{7!}{4!2!} \times 4! = 2520$
7(b)(ii)	$\uparrow _ \uparrow _ \uparrow _ \uparrow _ \uparrow _ \uparrow _ \uparrow$ Number of ways to arrange letters except "S" $= \frac{6!}{2!}$ Number of ways to choose arrows to slot in the "S" $= {}^7C_4$ Number of arrangements required $= \frac{6!}{2!} \times {}^7C_4 = 12600$
7(b)(iii)	E E $\uparrow _ \uparrow \quad \uparrow _ \uparrow _ \uparrow _ \uparrow$ Number of ways to arrange letters except "S" such that the two "E"s are together $= 5!$ Number of ways to choose arrows to slot in the "S" $= {}^6C_4$ Number of ways "S" to be separated and two "E"s are together $= 5! \times {}^6C_4 = 1800$ Number of arrangements required $= 12600 - 1800 = 10800$

Qn	Suggested Solution
8(a)(i)	 $x = 98 + 0.95y$ (join 2 points or $y = 1.0526x - 103.18$)
8(a)(ii)	Since the school wants to **estimate the laboratory light meter reading *x* from the phone app reading *y***, the error should be measured in the *x*-direction . Squaring residuals prevents **positive and negative residuals** from **cancelling out**.
8(a)(iii)	See part **(a)(i)**
8(a)(iv)	Sum of the squares of the residuals = 5349
8(b)(i)	$x = 1.0372y + 72.883$ (to 5 s.f.) $x = 1.04y + 72.9$ (to 3 s.f.) Using GC, $r = 0.906$
8(b)(ii)	Sum of the squares of the residuals $\approx 5172.43 < 5349$ (part **(a)(iv)**)
8(b)(iii)	Since $y = 300$ is **within the data range of [220,400]** (or **interpolation**), and $\lvert r \rvert = 0.906$ is **close to 1**, we would expect this value to be reliable.

Qn	Suggested Solution
9(a)	Let S be the mass of a Standard cheese block in kg. $S \sim \mathrm{N}\left(2.85, 0.2^2\right)$ $\mathrm{P}(S > 2.9) = 0.40129$ Let Y be the number of Standard cheese out of 90 with mass more than 2.9 kg. $Y \sim \mathrm{B}\left(90, 0.40129\right)$ Expected number $= \mathrm{E}(Y)$ $= 90 \times 0.40129$ $= 36.1161$ $= 36.1$ (3 s.f.)
9(b)	Let L be the mass of a Large cheese block in kg. $L \sim \mathrm{N}\left(4.25, 0.15^2\right)$ Let $D = L_1 + L_2 + ... + L_{10} - S_1 - S_2 - ..... - S_{15}$ $\mathrm{E}(D) = \mathrm{E}(L_1) + \mathrm{E}(L_2) + ... + \mathrm{E}(L_{10}) - \mathrm{E}(S_1) - \mathrm{E}(S_2) - .... - \mathrm{E}(S_{15})$ $= 10\left(4.25\right) - 15(2.85)$ $= -0.25$ $\mathrm{Var}(D) = \mathrm{Var}(L_1) + \mathrm{Var}(L_2) + ... + \mathrm{Var}(L_{10}) + \mathrm{Var}(S_1) + \mathrm{Var}(S_2) + ... + \mathrm{Var}(S_{15})$ $= 10\left(0.15^2\right) + 15(0.2^2)$ $= 0.825$ $D \sim \mathrm{N}\left(-0.25, 0.825\right)$ $\mathrm{P}(\lvert D \rvert < 2.3) = \mathrm{P}(-2.3 < D < 2.3)$ $= 0.9854984683$ $= 0.985$ (3 s.f.)
9(c)	Assume the masses of <u>all</u> cheese blocks are independent of one another.
9(d)	$\frac{L}{3} - \frac{S}{2} \sim \mathrm{N}\left(\frac{4.25}{3} - \frac{2.85}{2}, \frac{0.15^2}{3^2} + \frac{0.2^2}{2^2}\right)$ $\frac{L}{3} - \frac{S}{2} \sim \mathrm{N}\left(-0.0083333, 0.0125\right)$ $\mathrm{P}\left(\frac{L}{3} - \frac{S}{2} > 0\right) \approx 0.47029$ $= 0.470$ (3 s.f.)
9(e)	2 cuts are needed for the Large cheese block whereas only 1 cut is needed for the Standard cheese block. This means that there will be **double the amount of wastage for the Large cheese block.** This reduces the mean mass of Large Cheese portions more than Standard portions. $\mathrm{E}\left(\frac{L}{3} - \frac{S}{2}\right)$ will becomes more negative. $\mathrm{P}\left(\frac{L}{3} - \frac{S}{2} > 0\right)$ **should now be smaller than the one in part (d).**

Qn	Suggested Solution
10(a)	A random sample means that each order is selected independently from one another and each order has an equal chance of being selected.
10(b)	$\bar{x} = \dfrac{\sum(x-12)}{50} + 12 = 11.762$ $s^2 = \dfrac{1}{49}\left[17.36 - \dfrac{(-11.9)^2}{50}\right] \approx 0.29649 = 0.296$ (3 s.f)
10(c)	$H_0 : \mu = 12$ $H_1 : \mu < 12$ where μ is the **population mean** time taken Under H_0, since sample size of 50 is large, by Central Limit Theorem $\bar{X} \sim N\left(12, \dfrac{0.29649}{50}\right)$ approximately From GC, p-value $= P(\bar{X} \le 11.762) = 0.000998$ (3 s.f.) Since the p-value is very small, there is very strong evidence to reject H_0 and conclude that the company's claim is valid.
10(d)	$\bar{x} = k, \quad n = 80$ $s^2 = \dfrac{n}{n-1}(\text{sample variance}) = \dfrac{80}{79}(2.3^2)$ Under H_0, since $n = 80$ is large, by Central Limit Theorem $\bar{X} \sim N\left(12, \dfrac{2.3^2}{79}\right)$ approximately. Company's claim not supported $\Rightarrow$ we do not reject H_0 p-value > 0.05 $P(\bar{X} \le k) > 0.05$ From GC, $P(\bar{X} \le 11.57) = 0.05$ From the diagram, $k > 11.57$ (2 d.p) 
10(e)	As n is not large (< 30), we cannot apply Central Limit Theorem to approximate the distribution of $\bar{X}$. Hence, the test is not valid. We need to assume that the population preparation time is normally distributed.

Qn	Suggested Solution
11(ai)	$\mathrm{P}(X=4)$ $=\frac{3}{3}\times\frac{2}{3}\times\frac{1}{3}\times\frac{3}{3}$ $=\frac{2}{9}$ • 1st draw : any of the 3 colours • 2nd draw : any 2 colours different from the 1st draw • 3rd draw : the colour different from the 1st & 2nd draws • 4th draw : repeat any of the 3 colours drawn earlier $\mathrm{P}(X=2)$ $=\frac{3}{3}\times\frac{1}{3}=\frac{1}{3}$ • 1st draw : any of the 3 colours • 2nd draw : the same colour as the 1st draw $\mathrm{P}(X=3)$ $=\frac{3}{3}\times\frac{2}{3}\times\frac{2}{3}=\frac{4}{9}$ • 1st draw : any of the 3 colours • 2nd draw : any 2 colours different from the 1st draw • 3rd draw : repeat any of the 2 colours drawn earlier | x | 2 | 3 | 4 | | $\mathrm{P}(X=x)$ | $\frac{1}{3}=\frac{3}{9}$ | $\frac{4}{9}$ | $\frac{2}{9}$ |
11(aii)	$\mathrm{E}(X)=\left(2\times\frac{1}{3}\right)+\left(3\times\frac{4}{9}\right)+\left(4\times\frac{2}{9}\right)$ $=\frac{26}{9}$ $\mathrm{E}(X^2)=\left(2^2\times\frac{1}{3}\right)+\left(3^2\times\frac{4}{9}\right)+\left(4^2\times\frac{2}{9}\right)$ $=\frac{80}{9}$ $\mathrm{Var}(X)=\mathrm{E}(X^2)-[\mathrm{E}(X)]^2$ $=\frac{80}{9}-\left(\frac{26}{9}\right)^2$ $=\frac{44}{81}$
11(b)	Let Y be the number of blue marbles, out of 25 marbles. $Y\sim\mathrm{B}(25,\ 0.4)$ $\mathrm{P}(Y\le 12)=0.84623=0.846$ (3 sf)

11(c)	Let W be the number of boxes with <u>fewer blue marbles than non-blue ones</u>, out of 5 boxes. $W \sim \mathrm{B}(5,\ 0.84623)$ $\mathrm{P}(W \geq 3) = 1 - \mathrm{P}(W \leq 2)$ ≈ 0.97151 $= 0.972$ (3 sf)
11(d)	Let V be the number of blue marbles, out of $25 \times 5 = 125$ marbles. $V \sim \mathrm{B}(125,\ 0.4)$ $\mathrm{P}(V \leq 50) \approx 0.53875 = 0.539$ (3 sf)
11(e)	$Y \sim \mathrm{B}(25,\ 0.4)$ $\mathrm{E}(Y) = 25 \times 0.4 = 10$ $\mathrm{Var}(Y) = 25 \times 0.4 \times (1 - 0.4) = 6$ Since sample size = 30 is large, by Central Limit Theorem, $\overline{Y} = \frac{Y_1 + ... + Y_{30}}{30} \sim \mathrm{N}\left(10,\ \frac{6}{30}\right)$ approximately $\mathrm{P}(7 < \overline{Y} < 11) = 0.987$ (3 s.f.)